

WORKED KEY

Full Name: _____

Period: _____

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Evaluate $f(x) = 5 \cdot 3^x$. Find $f(2)$.
Answer: $5 \cdot 9 = 45$
2. Evaluate $f(x) = 24 \cdot 2^x$. Find $f(-3)$.
Answer: $24 \cdot (1/8) = 3$
3. Classify each function as exponential GROWTH or DECAY, and give the percent rate of change.
 - a. $y = 725(1.04)^x$
Answer: growth — 4% rate
 - b. $y = 560(0.65)^x$
Answer: decay — 35% rate
4. Write an exponential function to model the situation: a town of 4,000 residents grows 6% per year.
Answer: $y = 4000(1.06)^x$

5. A bakery sells 24 loaves of sourdough its first week, and weekly sales grow 25% each week. Predict the weekly sales after 3 weeks (round to the nearest loaf).
Answer: $24(1.25)^3 = 46.875 \rightarrow$ about 47 loaves
6. \$2,500 is invested at 3% interest, compounded annually. Find the balance after 4 years (nearest cent).
Answer: $2500(1.03)^4 \approx \$2,813.77$
7. A pond has 144 square feet of algae, and a treatment cuts the covered area in half each week. How much algae remains after 6 weeks?
Answer: $144(1/2)^6 = 2.25$ square feet

Algebra 1 – Module 9 Practice Test •

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8. Is the table linear or exponential? Give the function rule.

x	y
0	7
1	28
2	112
3	448

Answer: exponential, $\times 4$ each step: $y = 7 \cdot 4^x$

9. Plan L starts at \$400 and adds \$90 each year. Plan E starts at \$400 and grows 35% each year. Which is worth more after 4 years? Show both values.

Answer: L: $400 + 90(4) = \$760$; E: $400(1.35)^4 \approx \$1,328.60$ — the exponential plan

10. For the model $y = 480(1.22)^x$, identify the initial value and the factor, and interpret them.

Answer: initial value 480; growth factor 1.22, so it grows 22% each period

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11. Error Analysis — To find the value of 600 growing 15% per year for 2 years, Rex computed it by adding 30% at once: $600(1.30) = 780$. Identify his mistake and find the correct value.

Answer: percent growth COMPOUNDS — multiply by 1.15 twice: $600(1.15)^2 = \$793.50$

12. What value does $y = 9 \cdot 2^x$ approach as x becomes very negative, and what is that line called?

Answer: it approaches $y = 0$ — the asymptote

13. A social media post is seen by 8 people and the number who have seen it QUADRUPLES each hour. Write an exponential model, then find how many hours until at least 512 people have seen it.

Answer: $y = 8 \cdot 4^x$; $8 \cdot 4^3 = 512 \rightarrow 3$ hours

Cumulative Review — ACT Style. Circle the best answer.

14. Simplify: $\sqrt{48}$

(A) $16\sqrt{3}$
 (B) **$4\sqrt{3}$**
 (C) $3\sqrt{8}$
 (D) $6\sqrt{3}$

Answer: $\sqrt{(16 \cdot 3)} = 4\sqrt{3}$

15. Which ordered pair solves $y = x + 2$ and $x + y = 14$?

(E) (8, 6)
 (F) **(6, 8)**
 (G) (5, 9)
 (H) (7, 9)

Answer: $x + (x + 2) = 14 \rightarrow x = 6, y = 8$

16. Solve: $3(x + 4) = 5x - 2$

(A) 2
 (B) **7**
 (C) 10
 (D) -5

Answer: $3x + 12 = 5x - 2 \rightarrow 2x = 14 \rightarrow x = 7$